THE UNIVERSITY OF HONG KONG
DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE

STAT6011 COMPUTATIONAL STATISTICS AND BAYESIAN LEARNING
(2026 Fall)

Assignment 1, due on September 28

All computation MUST be conducted in R and the solution should be written using R markdown or latex (no handwritten text).

1. **Question 1**

   (a) Draw 10,000 random samples from the standard normal distribution $X \sim N(0,1)$. Let $Y = X^2$, and transform each sample to its square. Plot the histogram of Y samples in conjunction with the density of $\chi^2_{(1)}$.

   (b) Test whether these square-transformed samples are from a chi-squared distribution with 1 degree of freedom, $\chi^2_{(1)}$.

   [Total: 20 marks]

2. **Question 2**

   (a) To compute $\int_0^1 h(x)dx$, where

   $$h(x) = [\cos(50x) + \sin(20x)]^2,$$

   generate $u_1, \ldots, u_n$ $(n = 10,000)$ i.i.d. Unif$(0,1)$ random variables, and approximate $\int_0^1 h(x)dx = \int_0^1 h(x) \cdot 1dx$ with $\sum_{i=1}^n h(u_i)/n$.

   (b) It is clear that any $X \sim$ Beta(α, β) density function $f(x)$ can be used to simulate $u_1, \ldots, u_n$ $(n = 10,000)$, because

   $$\int_0^1 h(x)dx = \int_0^1 \frac{h(x)}{f(x)} f(x)dx.$$

   Find the optimal values of Beta distribution α and β to achieve the highest efficiency for the Monte Carlo approximation (i.e., with $n = 10,000$ the approximation would be the most accurate).

   (c) Use Gaussian quadrature to compute the integral $\int_0^1 h(x)dx$, which is treated as the benchmark for comparison.

   [Total: 30 marks]

3. **Question 3**

(a) Fit a linear model to the data in Q3.csv,

$$y = \boldsymbol{\beta}^\top \mathbf{X} + \epsilon,$$

where $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_{10})^\top$ is a vector of regression coefficients and the error $\epsilon \sim N(0, \sigma^2)$. We assign each β_j $(j = 1, \ldots, 11)$ a normal $N(0, 5^2)$ prior and σ^2 an inverse-gamma(0.1, 0.1) prior.

Apply rejection sampling to draw posterior samples. Let $\boldsymbol{\theta} = (\boldsymbol{\beta}, \sigma^2)$. Using the prior $\pi(\boldsymbol{\theta})$ as the candidate, compute

$$C^* = \sup_{\boldsymbol{\theta} \in \Theta} \frac{\pi(\boldsymbol{\theta}|\boldsymbol{y})}{\pi(\boldsymbol{\theta})} \propto \sup_{\boldsymbol{\theta} \in \Theta} f(\boldsymbol{y}|\boldsymbol{\theta}) = f(\boldsymbol{y}|\hat{\boldsymbol{\theta}}),$$

where $\hat{\boldsymbol{\theta}}$ is the MLE.

Implement the rejection sampling as follows:

- Simulate $u \sim \text{Unif}[0, 1]$,
- Simulate $\boldsymbol{\theta} \sim \pi(\boldsymbol{\theta})$,
- Accept $\boldsymbol{\theta}$ if $u \leq \frac{\pi(\boldsymbol{\theta}|\boldsymbol{y})}{C^* \pi(\boldsymbol{\theta})} = \frac{f(\boldsymbol{y}|\boldsymbol{\theta})}{f(\boldsymbol{y}|\hat{\boldsymbol{\theta}})}$.

Draw 1000 posterior samples of $\boldsymbol{\beta}, \sigma^2$, plot the histograms and densities, and compute the posterior means and variances. Compare them with the MLEs.

(b) Change the prior of each β_j $(j = 1, \ldots, 11)$ to be Laplace$(0, 1)$ with density

$$f(\beta) = \frac{1}{2} \exp(-|\beta|),$$

and repeat the computation. Which prior is more efficient (i.e., with fewer rejections)?

(c) Some of the true values of β are zero (sparsity), which should be shrunk to zero. The normal prior on β can be viewed as ridge regression. Fit a ridge regression model and compare the results. The Laplace prior on β can be viewed as Lasso regression. Fit a Lasso regression model and compare the results.

[Total: 30 marks]

4. **Question 4 (Prior-posterior mode algorithm)**

In the lecture, we showed that the posterior distribution of θ is

$$p(\theta|\mathbf{y}) \propto p(\mathbf{y}|\theta) p(\theta),$$

where $p(\mathbf{y}|\theta)$ is the likelihood and $p(\theta)$ is a prior. By taking the logarithm on both sides, we have

$$\log p(\theta|\mathbf{y}) \propto \log p(\mathbf{y}|\theta) + \log p(\theta).$$

Suppose that θ^* is the posterior mode as well as the prior mode, and then we take the first derivative and evaluate it at θ^*

$$\frac{\partial \log p(\theta^*|\mathbf{y})}{\partial \theta} \propto \frac{\partial \log p(\mathbf{y}|\theta^*)}{\partial \theta} + \frac{\partial \log p(\theta^*)}{\partial \theta}.$$

Because both $\frac{\partial \log p(\theta^*|\mathbf{y})}{\partial \theta} = 0$ and $\frac{\partial \log p(\theta^*)}{\partial \theta} = 0$ due to θ^* being the maximum (or mode) of the posterior and the prior. So $\frac{\partial \log p(\mathbf{y}|\theta^*)}{\partial \theta} = 0$, and thus θ^* is the MLE.

Read Y from Q4.csv. Assume $Y \sim N(\mu, \sigma^2)$ where $\sigma = 1$ is known. Assign a normal prior to the mean, $\mu \sim N(0, 2^2)$.

(a) Derive the posterior distribution of μ.

(b) Design the PPMM (prior-posterior mode matching) algorithm that iteratively updates the prior mode (or mean) using the posterior mode (or mean) till convergence. That is to iterate the following two steps till convergence:

   (i) At step j, draw 1000 posterior samples of μ, compute the posterior mean $\hat{\mu}_{[j]}$.

   (ii) At step $j+1$, set the new prior mean of μ at the obtained posterior mean in the jth step, $\mu \sim N(\hat{\mu}_{[j]}, 2^2)$. Go back to (i), and repeat till convergence.

   Apply PPMM to find the MLE for μ, and compare the PPMM estimate with the traditional MLE (sample mean).

[Total: 20 marks]